

MAT1001 Calculus for Engineers

Introduction to Vector Calculus



VIT-AP
UNIVERSITY

Dr. Aswathy R K

Department of Mathematics

School of Advanced Sciences

Vellore Institute of Technology – Andhra Pradesh

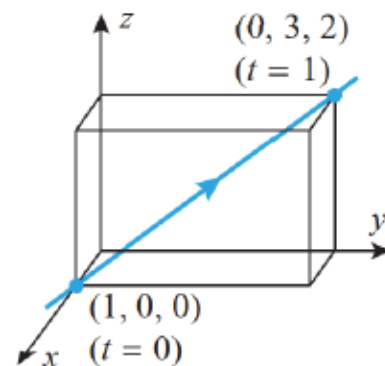
Parametric Curve

- ▶ If f , g , and h are three well-behaved functions, then the parametric equations

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

generate a curve in 3-space that is traced in a specific direction as t increases, this direction is called the orientation or direction of increasing parameter.

- ▶ The curve together with its orientation is called the graph of the parametric equations or the parametric curve represented by the equations.
- ▶ If no restrictions are stated explicitly or are implied by the equations, then it will be understood that t varies over the interval $(-\infty, +\infty)$.
- ▶ **Example** The parametric equations $x = 1 - t$, $y = 3t$, $z = 2t$ represent a line in 3-space that passes through the point $(1, 0, 0)$ and is parallel to the vector $\langle -1, 3, 2 \rangle$. Since x decreases as t increases, the line has the orientation shown in figure



$$x = 1 - t, \quad y = 3t, \quad z = 2t$$

Vector Valued Functions

- ▶ For each value of t , we can view $\langle x(t), y(t), z(t) \rangle$ as a terminal point for a vector $\mathbf{r}$ whose initial point is origin, that is,

$$\mathbf{r} = \mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

then we obtain $\mathbf{r}$ as a function of the parameter t , that is, $\mathbf{r} = \mathbf{r}(t)$.

Since this function produces a *vector*, we say that $\mathbf{r} = \mathbf{r}(t)$ defines $\mathbf{r}$ as a vector-valued function of a real variable, or more simply, a vector-valued function.

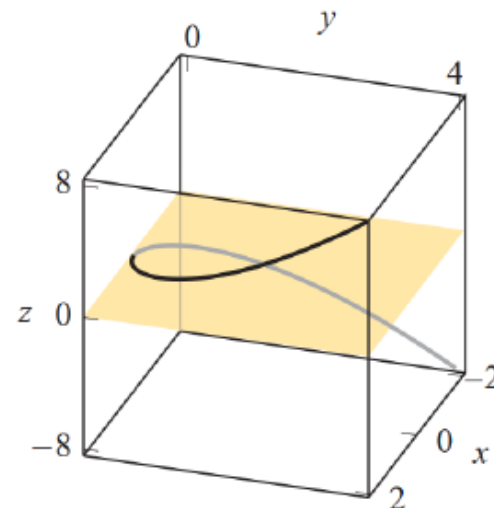
- ▶ The functions $x(t)$, $y(t)$, and $z(t)$ are called the component functions or the components of $\mathbf{r}(t)$.

- ▶ **Example** The component functions of

$$\mathbf{r}(t) = \langle t, t^2, t^3 \rangle = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$$

are

$$x(t) = t, \quad y(t) = t^2, \quad z(t) = t^3$$



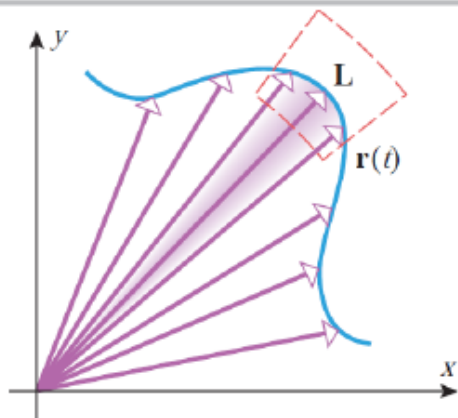
Limit of a Vector Valued Function

DEFINITION Let $\mathbf{r}(t)$ be a vector-valued function that is defined for all t in some open interval containing the number a , except that $\mathbf{r}(t)$ need not be defined at a . We will write

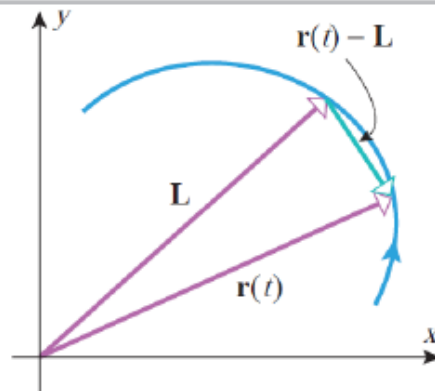
$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{L}$$

if and only if

$$\lim_{t \rightarrow a} \|\mathbf{r}(t) - \mathbf{L}\| = 0$$



$\mathbf{r}(t)$ approaches $\mathbf{L}$ in length and direction if $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{L}$.



$\|\mathbf{r}(t) - \mathbf{L}\|$ is the distance between terminal points for vectors $\mathbf{r}(t)$ and $\mathbf{L}$ when positioned with the same initial points.

Limit of a Vector Valued Function

THEOREM

(a) If $\mathbf{r}(t) = \langle x(t), y(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j}$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} x(t), \lim_{t \rightarrow a} y(t) \right\rangle = \lim_{t \rightarrow a} x(t)\mathbf{i} + \lim_{t \rightarrow a} y(t)\mathbf{j}$$

provided the limits of the component functions exist. Conversely, the limits of the component functions exist provided $\mathbf{r}(t)$ approaches a limiting vector as t approaches a .

(b) If $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, then

$$\begin{aligned} \lim_{t \rightarrow a} \mathbf{r}(t) &= \left\langle \lim_{t \rightarrow a} x(t), \lim_{t \rightarrow a} y(t), \lim_{t \rightarrow a} z(t) \right\rangle \\ &= \lim_{t \rightarrow a} x(t)\mathbf{i} + \lim_{t \rightarrow a} y(t)\mathbf{j} + \lim_{t \rightarrow a} z(t)\mathbf{k} \end{aligned}$$

provided the limits of the component functions exist. Conversely, the limits of the component functions exist provided $\mathbf{r}(t)$ approaches a limiting vector as t approaches a .

► **Example** Let $\mathbf{r}(t) = t^2\mathbf{i} + e^t\mathbf{j} - (2 \cos \pi t)\mathbf{k}$. Then

$$\lim_{t \rightarrow 0} \mathbf{r}(t) = \left(\lim_{t \rightarrow 0} t^2 \right) \mathbf{i} + \left(\lim_{t \rightarrow 0} e^t \right) \mathbf{j} - \left(\lim_{t \rightarrow 0} 2 \cos \pi t \right) \mathbf{k} = \mathbf{j} - 2\mathbf{k}$$

Alternatively, using the angle bracket notation for vectors,

$$\lim_{t \rightarrow 0} \mathbf{r}(t) = \lim_{t \rightarrow 0} \langle t^2, e^t, -2 \cos \pi t \rangle = \left\langle \lim_{t \rightarrow 0} t^2, \lim_{t \rightarrow 0} e^t, \lim_{t \rightarrow 0} (-2 \cos \pi t) \right\rangle = \langle 0, 1, -2 \rangle$$

DEFINITION A vector-valued function $\mathbf{r}(t)$ to be *continuous* at $t = a$ if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$$

That is, $\mathbf{r}(a)$ is defined, the limit of $\mathbf{r}(t)$ as $t \rightarrow a$ exists, and the two are equal.

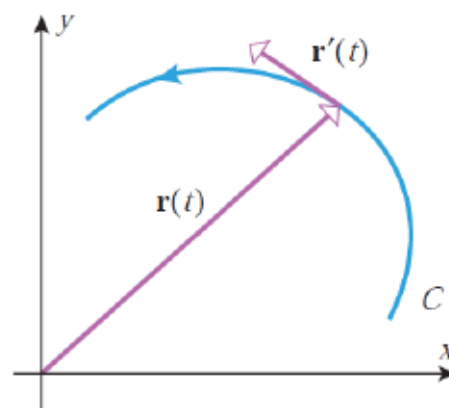
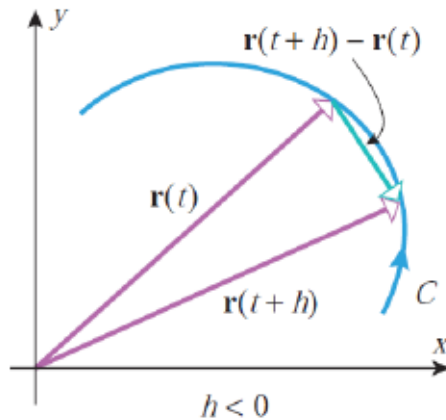
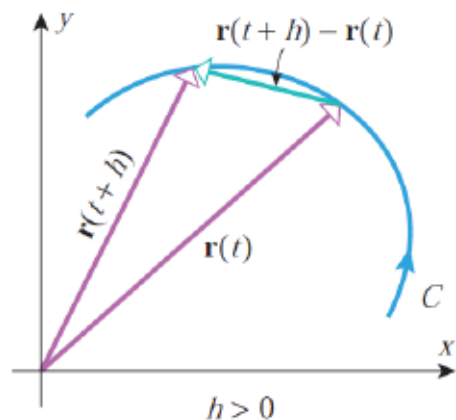
A vector-valued function is continuous at $t = a$ if and only if its component functions are continuous at $t = a$.

Differentiability of a Vector Valued Function

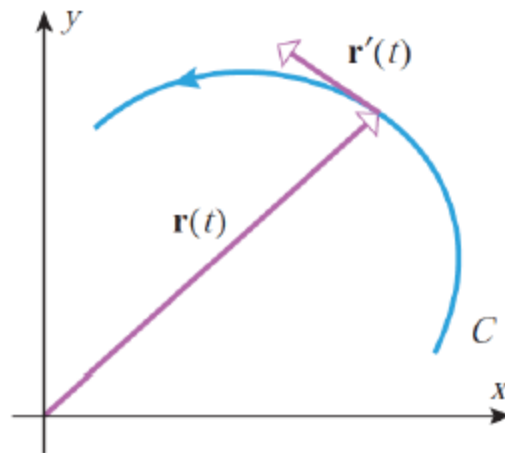
DEFINITION If $\mathbf{r}(t)$ is a vector-valued function, we define the *derivative of $\mathbf{r}$ with respect to t* to be the vector-valued function $\mathbf{r}'$ given by

$$\frac{d}{dt}[\mathbf{r}(t)] = \frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

The domain of $\mathbf{r}'$ consists of all values of t in the domain of $\mathbf{r}(t)$ for which the limit exists.



GEOMETRIC INTERPRETATION OF THE DERIVATIVE Suppose that C is the graph of a vector-valued function $\mathbf{r}(t)$ in 2-space or 3-space and that $\mathbf{r}'(t)$ exists and is nonzero for a given value of t . If the vector $\mathbf{r}'(t)$ is positioned with its initial point at the terminal point of the radius vector $\mathbf{r}(t)$, then $\mathbf{r}'(t)$ is tangent to C and points in the direction of increasing parameter.



THEOREM *If $\mathbf{r}(t)$ is a vector-valued function, then $\mathbf{r}$ is differentiable at t if and only if each of its component functions is differentiable at t , in which case the component functions of $\mathbf{r}'(t)$ are the derivatives of the corresponding component functions of $\mathbf{r}(t)$.*

Assume that $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Then $\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$

► **Example** Let $\mathbf{r}(t) = t^2\mathbf{i} + e^t\mathbf{j} - (2 \cos \pi t)\mathbf{k}$. Then

$$\begin{aligned}\mathbf{r}'(t) &= \frac{d}{dt}(t^2)\mathbf{i} + \frac{d}{dt}(e^t)\mathbf{j} - \frac{d}{dt}(2 \cos \pi t)\mathbf{k} \\ &= 2t\mathbf{i} + e^t\mathbf{j} + (2\pi \sin \pi t)\mathbf{k}\end{aligned}$$

THEOREM (Rules of Differentiation) Let $\mathbf{r}(t)$, $\mathbf{r}_1(t)$, and $\mathbf{r}_2(t)$ be differentiable vector-valued functions that are all in 2-space or all in 3-space, and let $f(t)$ be a differentiable real-valued function, k a scalar, and $\mathbf{c}$ a constant vector (that is, a vector whose value does not depend on t). Then the following rules of differentiation hold:

$$(a) \quad \frac{d}{dt}[\mathbf{c}] = \mathbf{0}$$

$$(b) \quad \frac{d}{dt}[k\mathbf{r}(t)] = k \frac{d}{dt}[\mathbf{r}(t)]$$

$$(c) \quad \frac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] + \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$(d) \quad \frac{d}{dt}[\mathbf{r}_1(t) - \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] - \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$(e) \quad \frac{d}{dt}[f(t)\mathbf{r}(t)] = f(t) \frac{d}{dt}[\mathbf{r}(t)] + \frac{d}{dt}[f(t)]\mathbf{r}(t)$$

THEOREM If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t , then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t .

► Example

Let $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j}$

It is clear that $\|\mathbf{r}(t)\| = \sqrt{\cos(t)^2 + \sin(t)^2} = 1 \quad \forall t$

and $\mathbf{r}'(t) = -\sin(t)\mathbf{i} + \cos(t)\mathbf{j}$

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = -\sin(t)\cos(t) + \sin(t)\cos(t) = 0$$

Smooth Parameterization

DEFINITION A curve C is represented by $\mathbf{r}(t)$ is said to be **smoothly parameterized** by $\mathbf{r}(t)$ (that is $\mathbf{r}(t)$ is a smooth function of t) if $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$ for any allowable value of t . Geometrically, this means that a smoothly parametrized curve can have no abrupt changes in direction as the parameter increases.

► **Example** Determine whether the following vector-valued functions are smooth.

(a) $\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j} + ct \mathbf{k} \quad (a > 0, c > 0)$

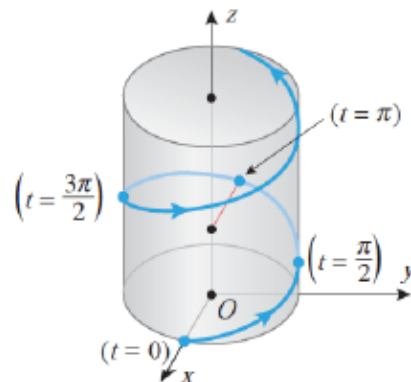
(b) $\mathbf{r}(t) = t^2 \mathbf{i} + t^3 \mathbf{j}$

Solution (a).

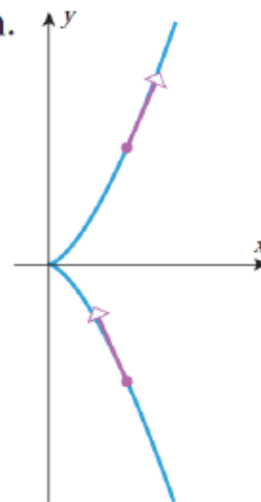
$$\mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + c \mathbf{k}$$

Solution (b).

$$\mathbf{r}'(t) = 2t \mathbf{i} + 3t^2 \mathbf{j}$$



$$x = a \cos t, \quad y = a \sin t, \quad z = ct$$

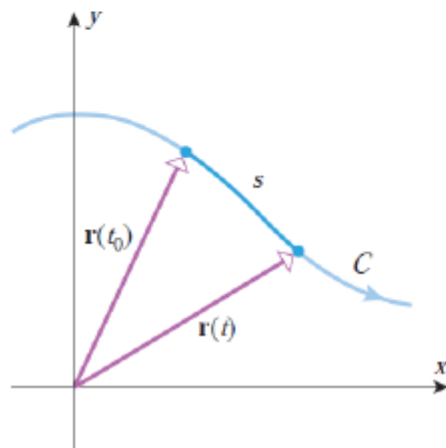


$$\mathbf{r}'(t) = t^2 \mathbf{i} + t^3 \mathbf{j}$$

Arc Length Parameter

THEOREM Let C be the graph of a smooth vector-valued function $\mathbf{r}(t)$ in 2-space or 3-space, and let $\mathbf{r}(t_0)$ be any point on C . Then the following formula defines a positive change of parameter from t to s , where s is an arc length parameter having $\mathbf{r}(t_0)$ as its reference point

$$s = \int_{t_0}^t \left\| \frac{d\mathbf{r}}{du} \right\| du \quad (10)$$



Example

► **Example** Find the arc length parametrization of the circular helix

$$\mathbf{r} = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k} \quad (13)$$

that has reference point $\mathbf{r}(0) = (1, 0, 0)$ and the same orientation as the given helix.

Solution. Replacing t by u in $\mathbf{r}$ for integration purposes and taking $t_0 = 0$ in Formula (10), we obtain

$$\mathbf{r} = \cos u \mathbf{i} + \sin u \mathbf{j} + u \mathbf{k} \quad \text{and} \quad \frac{d\mathbf{r}}{du} = (-\sin u) \mathbf{i} + \cos u \mathbf{j} + \mathbf{k}$$

$$\left\| \frac{d\mathbf{r}}{du} \right\| = \sqrt{(-\sin u)^2 + \cos^2 u + 1} = \sqrt{2} \quad \text{and} \quad s = \int_0^t \left\| \frac{d\mathbf{r}}{du} \right\| du = \int_0^t \sqrt{2} du = \sqrt{2}u \Big|_0^t = \sqrt{2}t$$

Thus, $t = s/\sqrt{2}$, so (13) can be reparametrized in terms of s as

$$\mathbf{r} = \cos \left(\frac{s}{\sqrt{2}} \right) \mathbf{i} + \sin \left(\frac{s}{\sqrt{2}} \right) \mathbf{j} + \frac{s}{\sqrt{2}} \mathbf{k}$$

We are guaranteed that this reparametrization preserves the orientation of the helix since Formula (10) produces a positive change of parameter.

Properties of Arc Length Parameterization

THEOREM

- (a) If C is the graph of a smooth vector-valued function $\mathbf{r}(t)$ in 2-space or 3-space, where t is a general parameter, and if s is the arc length parameter for C defined by Formula (10), then for every value of t the tangent vector has length

$$\left\| \frac{d\mathbf{r}}{dt} \right\| = \frac{ds}{dt}$$

- (b) If C is the graph of a smooth vector-valued function $\mathbf{r}(s)$ in 2-space or 3-space, where s is an arc length parameter, then for every value of s the tangent vector to C has length

$$\left\| \frac{d\mathbf{r}}{ds} \right\| = 1$$

- (c) If C is the graph of a smooth vector-valued function $\mathbf{r}(t)$ in 2-space or 3-space, and if $\|d\mathbf{r}/dt\| = 1$ for every value of t , then for any value of t_0 in the domain of $\mathbf{r}$, the parameter $s = t - t_0$ is an arc length parameter that has its reference point at the point on C where $t = t_0$.

A large, faint watermark of the VIT logo is centered in the background of the slide. It is the same circular emblem as seen in the header, featuring a temple gopuram and the text "VIT" and "VELLORE INSTITUTE OF TECHNOLOGY".

Thank You...

VIT-AP
UNIVERSITY